

Course: Cloud and Network Security – CII - 2026

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Sunday, September 20, 2026

Week 2 Assignment 1:

HTB Academy: Introduction to Network Traffic Analysis

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Introduction

This activity will involve using Hack the Box to perform Network Traffic Analysis (NTA). This can be described as the act of examining network traffic to characterize common ports and protocols utilized, establish a baseline for our environment, monitor and respond to threats, and ensure the greatest possible insight into our organization's network. For the activity, we are going to use Wireshark. We shall put to practice the NTA Workflow (Ingest Traffic>Reduce Noise by Filtering>Analyze and Explore>Detect and Alert). We'll also interact with tcpdump, tshark and termshark, familiarize with the structure as well as common switches.

Answers to questions: Introduction to Network Traffic Analysis

Module 1: Introduction

Section: Networking Primer - Layers 1-4

Question 1

How many layers does the OSI model have?

7

Question 2

How many layers are there in the TCP/IP model?

4

Question 3

True or False: Routers operate at layer 2 of the OSI model?

False

Question 4

What addressing mechanism is used at the Link Layer of the TCP/IP model?

MAC-address

Question 5

At what layer of the OSI model is a PDU encapsulated into a packet? (the number)

3

Question 6

What addressing mechanism utilizes a 32-bit address?

IPv4

Week 2 Assignment 1 HTB Academy: Introduction to Network Traffic Analysis

Question 7

What Transport layer protocol is connection oriented?

TCP

Question 8

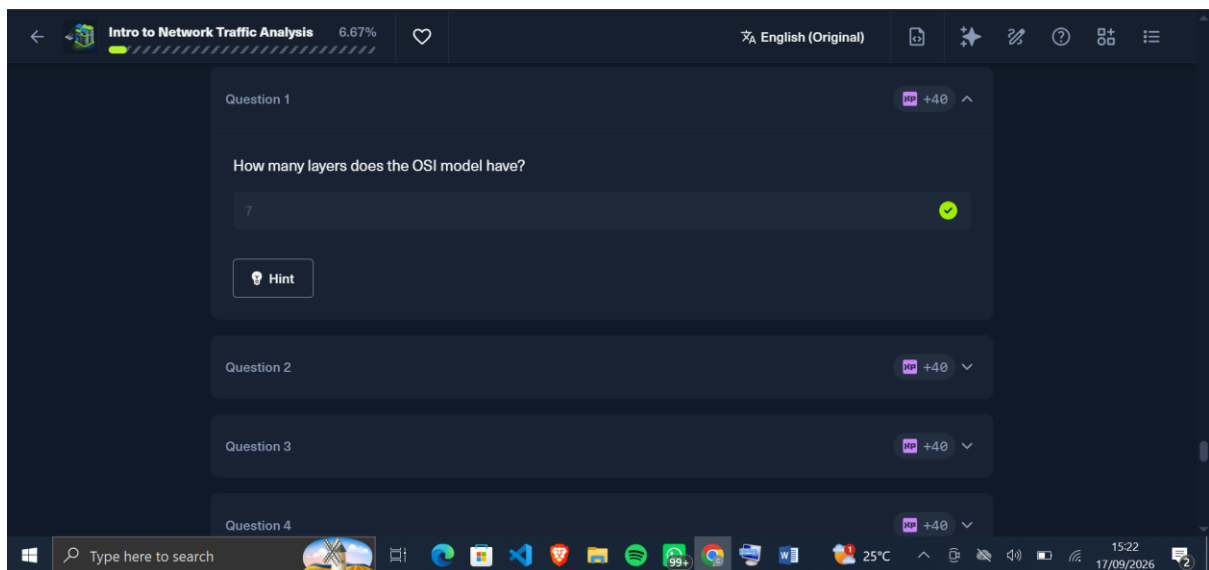
What Transport Layer protocol is considered unreliable?

UDP

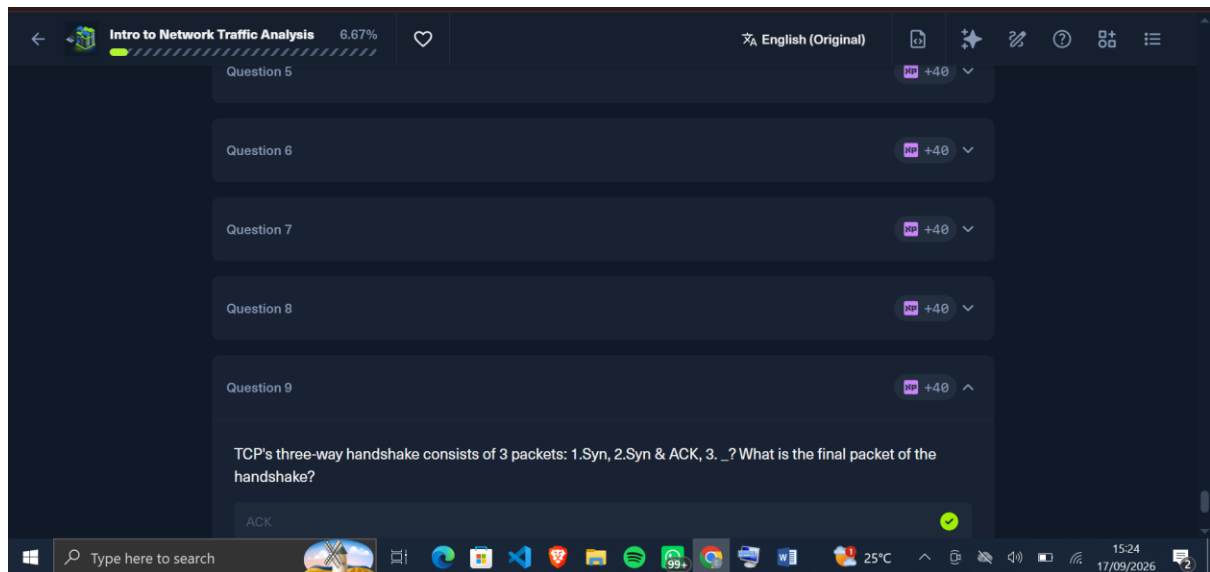
Question 9

TCP's three-way handshake consists of 3 packets: 1.Syn, 2.Syn & ACK, 3. _? What is the final packet of the handshake?

ACK



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Section: Networking Primer - Layers 5-7

Question 1

What is the default operational mode method used by FTP?

Active

Question 2

FTP utilizes what two ports for command and data transfer? (separate the two numbers with a space)

20 21

Question 3

Does SMB utilize TCP or UDP as its transport layer protocol?

TCP

Question 4

SMB has moved to using what TCP port?

445

Question 5

Hypertext Transfer Protocol uses what well known TCP port number?

80

Question 6

What HTTP method is used to request information and content from the webserver?

GET

Question 7

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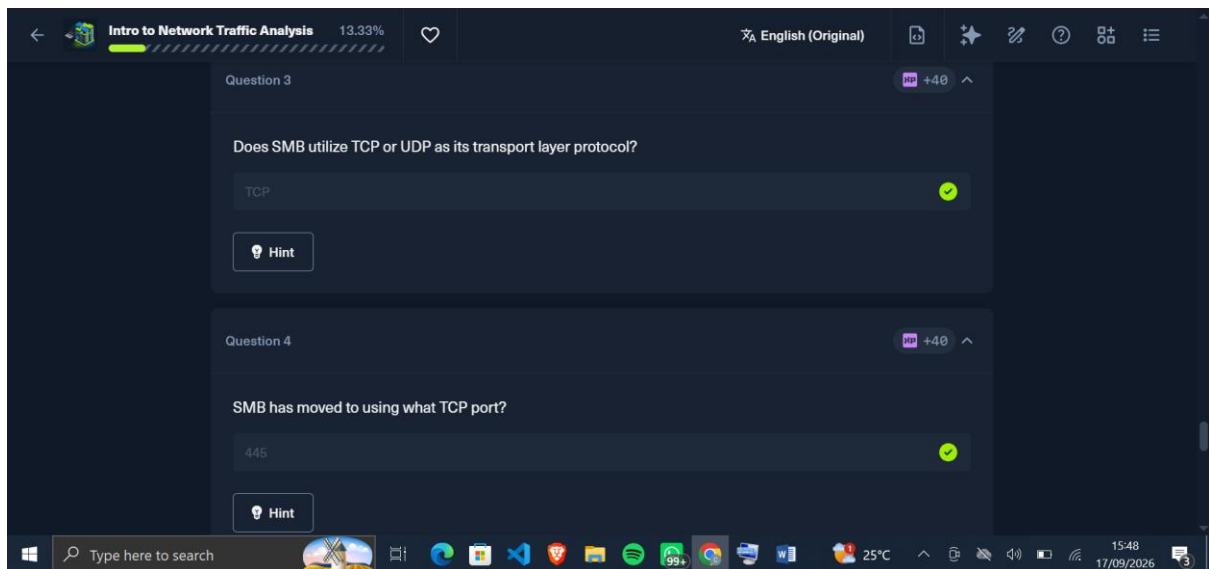
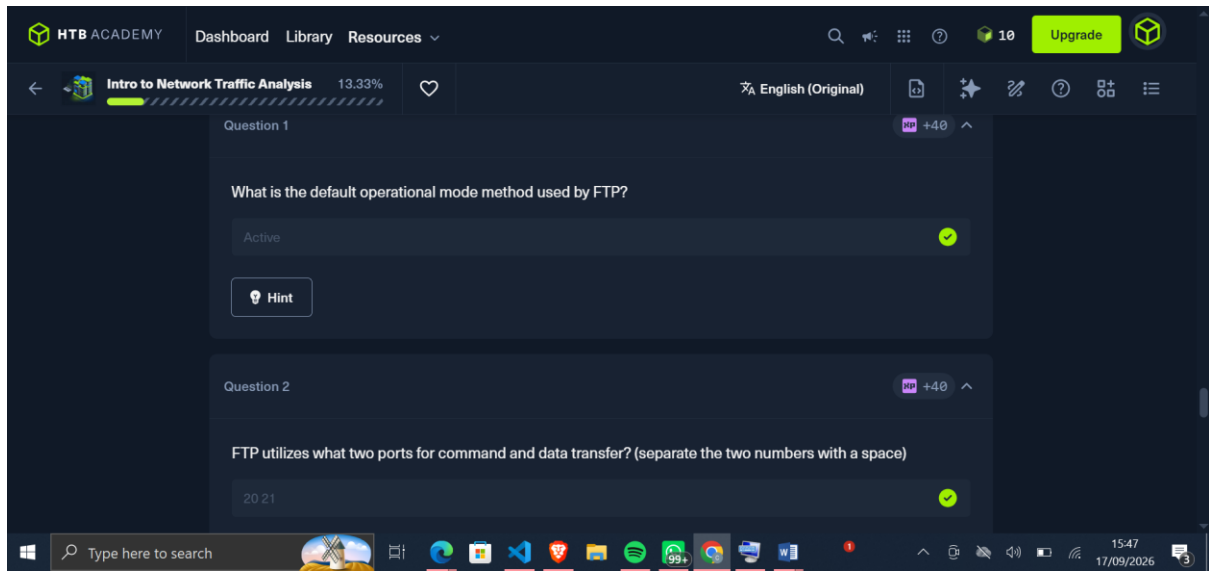
What web based protocol uses TLS as a security measure?

HTTPS

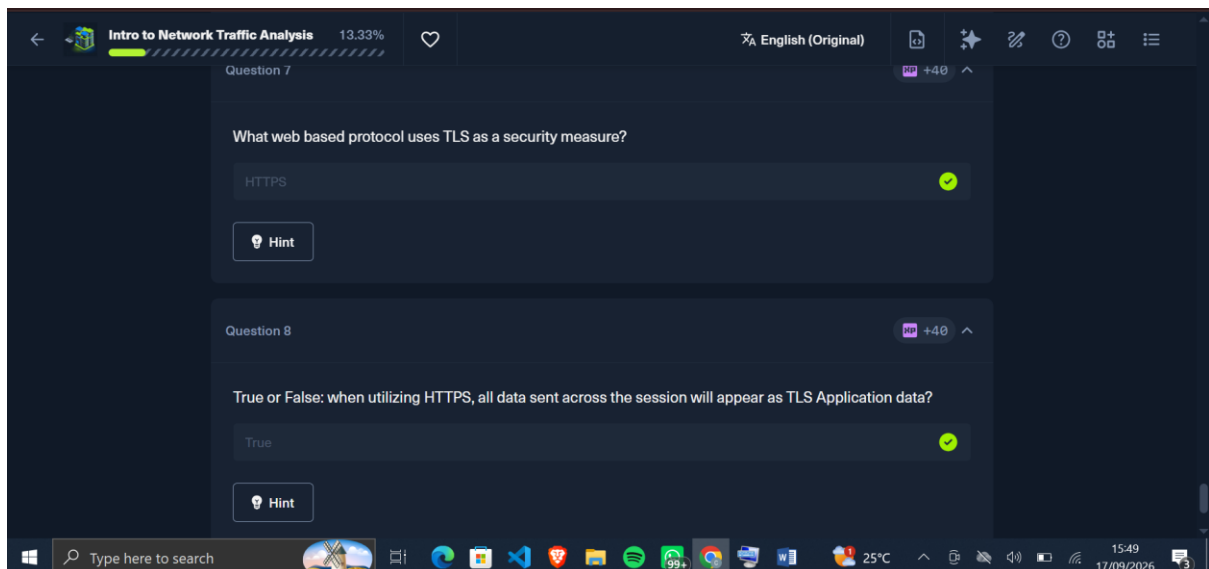
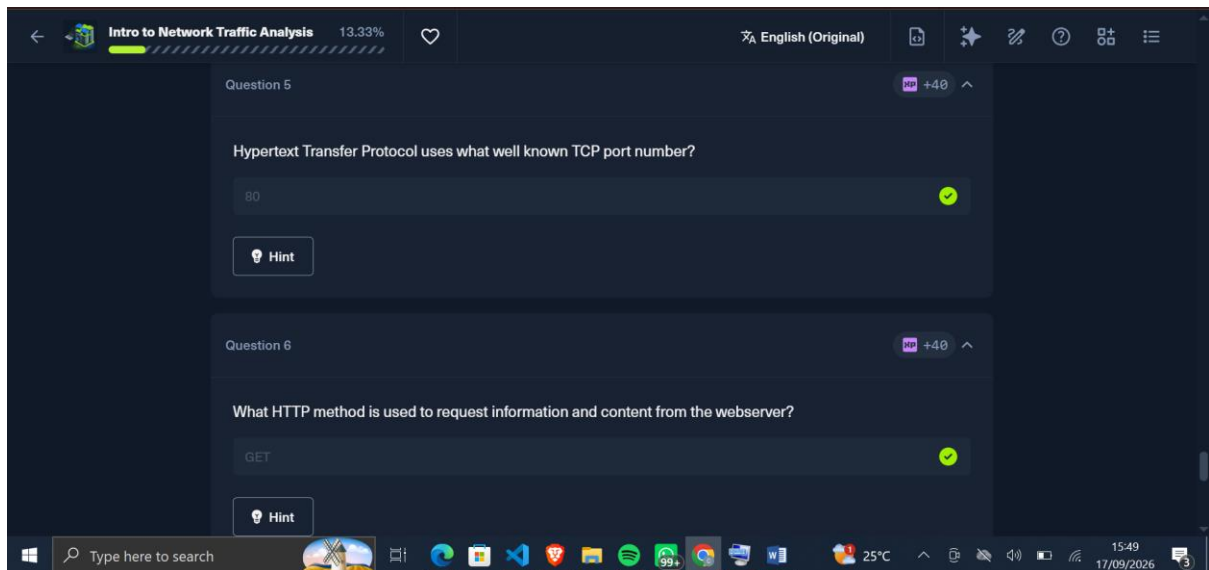
Question 8

True or False: when utilizing HTTPS, all data sent across the session will appear as TLS Application data?

True



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Note: Module 2 has no questions

Module 3: TCPDUMP

Section: Tcpdump Fundamentals

Question 1

Utilizing the output shown in question-1.png, who is the server in this communication? (IP Address)

174.143.213.184

Question 2

Were absolute or relative sequence numbers used during the capture? (see question-1.zip to answer)

relative

Question 3

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If I wish to start a capture without hostname resolution, verbose output, showing contents in ASCII and hex, and grab the first 100 packets; what are the switches used? please answer in the order the switches are asked for in the question.

`-nvXc 100`

Question 4

Given the capture file at /tmp/capture.pcap, what tcpdump command will enable you to read from the capture and show the output contents in Hex and ASCII? (Please use best practices when using switches)

`sudo tcpdump -Xr /tmp/capture.pcap`

Question 5

What TCPDump switch will increase the verbosity of our output? (Include the - with the proper switch)

`-v`

Question 6

What built in terminal help reference can tell us more about TCPDump?

`man`

Question 7

What TCPDump switch will let me write my output to a file?

`-W`

The screenshot shows the HTB Academy interface for a quiz titled "Intro to Network Traffic Analysis". The progress bar indicates 33.33% completion. The quiz contains two questions:

- Question 1:** "Utilizing the output shown in question-1.png, who is the server in this communication? (IP Address)". The answer "174.143.213.184" is entered and marked correct with a green checkmark. There are "Hint" and "Download File" buttons.
- Question 2:** "Were absolute or relative sequence numbers used during the capture? (see question-1.zip to answer)". The answer "relative" is entered and marked correct with a green checkmark.

On the right side, there is a "Table of Contents" panel showing the following sections:

- 2 Analysis (checked)
- 3 Tcpdump (4 Sections)
 - Interactive
 - Tcpdump Fundamentals (In progress)
 - Interactive
 - Capturing With Tcpdump (Fundamentals Labs)
 - Interactive
 - Tcpdump Packet Filtering

The bottom of the screen shows a Windows taskbar with the search bar, task icons, and system tray showing the date and time as 16:53 on 19/09/2026.

Week 2 Assignment 1 HTB Academy: Introduction to Network Traffic Analysis

Intro to Network Traffic Analysis 33.33%

English (Original)

Table of Contents

Question 3

If I wish to start a capture without hostname resolution, verbose output, showing contents in ASCII and hex, and grab the first 100 packets; what are the switches used? please answer in the order the switches are asked for in the question.

`-nvXc 100`

Hint

Question 4

Given the capture file at /tmp/capture.pcap, what tcpdump command will enable you to read from the capture and show the output contents in Hex and ASCII? (Please use best practices when using switches)

`sudo tcpdump -Xr /tmp/capture.pcap`

Table of Contents

2 Analysis

3 Tcpdump 4 Sections

Interactive Tcpdump Fundamentals In progress

Interactive Capturing With Tcpdump (Fundamentals Labs)

Interactive Tcpdump Packet Filtering

Type here to search

26°C 16:54 19/09/2026

Intro to Network Traffic Analysis 33.33%

English (Original)

Table of Contents

Question 5

What TCPDump switch will increase the verbosity of our output? (Include the - with the proper switch)

`-v`

Hint

Question 6

What built in terminal help reference can tell us more about TCPDump?

`man`

Hint

Table of Contents

2 Analysis

3 Tcpdump 4 Sections

Interactive Tcpdump Fundamentals In progress

Interactive Capturing With Tcpdump (Fundamentals Labs)

Interactive Tcpdump Packet Filtering

Type here to search

26°C 16:55 19/09/2026

Intro to Network Traffic Analysis 33.33%

English (Original)

Table of Contents

Question 7

What TCPDump switch will let me write my output to a file?

`-w`

Hint

Table of Contents

2 Analysis

3 Tcpdump 4 Sections

Interactive Tcpdump Fundamentals In progress

Interactive Capturing With Tcpdump (Fundamentals Labs)

Previous

Section 6 / 15

Next

Mark Complete & Next

Type here to search

26°C 16:55 19/09/2026

Week 2 Assignment 1 HTB Academy: Introduction to Network Traffic Analysis

Section: Fundamentals Lab

Question 1

What TCPDump switch will allow us to pipe the contents of a pcap file out to another function such as 'grep'?

-l

Question 2

True or False: The filter "port" looks at source and destination traffic.

true

Question 3

If we wished to filter out ICMP traffic from our capture, what filter could we use? (word only, not symbol please.)

not icmp

Question 4

What command will show you where / if TCPDump is installed?

which tcpdump

Question 5

How do you start a capture with TCPDump to capture on eth0?

tcpdump -i eth0

Question 6

What switch will provide more verbosity in your output?

-v

Question 7

What switch will write your capture output to a .pcap file?

-w

Question 8

What switch will read a capture from a .pcap file?

-r

Question 9

What switch will show the contents of a capture in Hex and ASCII?

-X

Week 2 Assignment 1 HTB Academy: Introduction to Network Traffic Analysis

Intro to Network Traffic Analysis 40%

English (Original)

What TCPDump switch will allow us to pipe the contents of a pcap file out to another function such as 'grep'?

-l

Hint

Question 2 +40

Question 3 +40

If we wished to filter out ICMP traffic from our capture, what filter could we use? (word only, not symbol please.)

not icmp

Type here to search

24°C 18:20 19/09/2026

Intro to Network Traffic Analysis 40%

English (Original)

Question 4 +40

What command will show you where / if TCPDump is installed?

which tcpdump

Hint

Question 5 +40

Question 6 +40

What switch will provide more verbosity in your output?

-v

Type here to search

23°C 18:50 19/09/2026

Intro to Network Traffic Analysis 40%

English (Original)

Question 7 +40

What switch will write your capture output to a .pcap file?

-w

Hint

Question 8 +40

Question 9 +40

What switch will show the contents of a capture in Hex and ASCII?

-X

Type here to search

23°C 18:51 19/09/2026

Week 2 Assignment 1 HTB Academy: Introduction to Network Traffic Analysis

Section: Tcpdump Packet Filtering

Question 1

What filter will allow me to see traffic coming from or destined to the host with an ip of 10.10.20.1?

host 10.10.20.1

Question 2

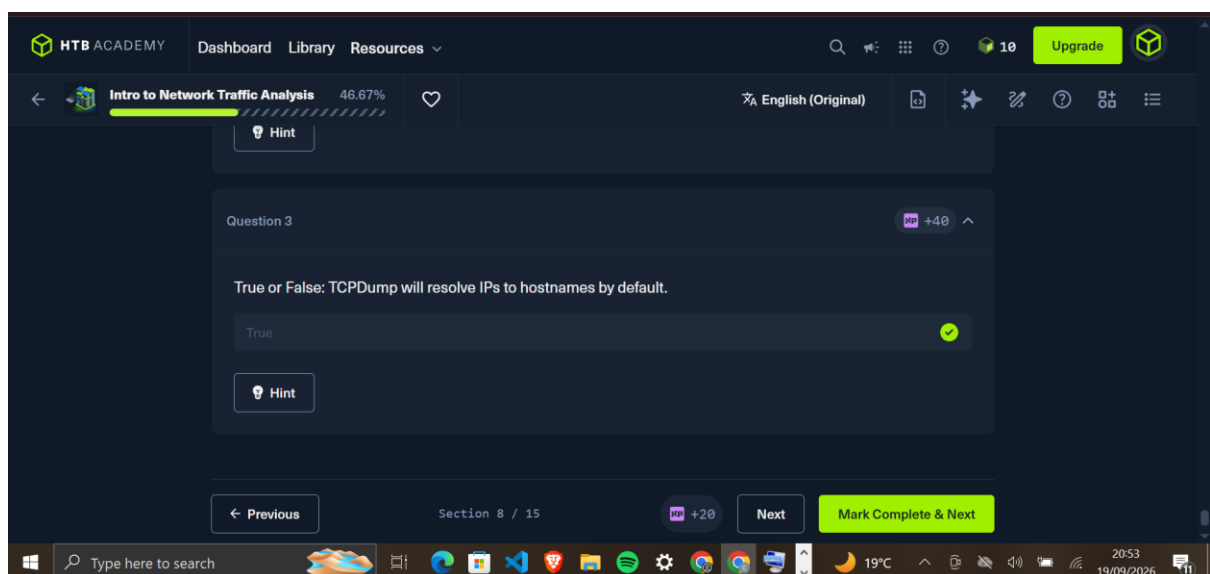
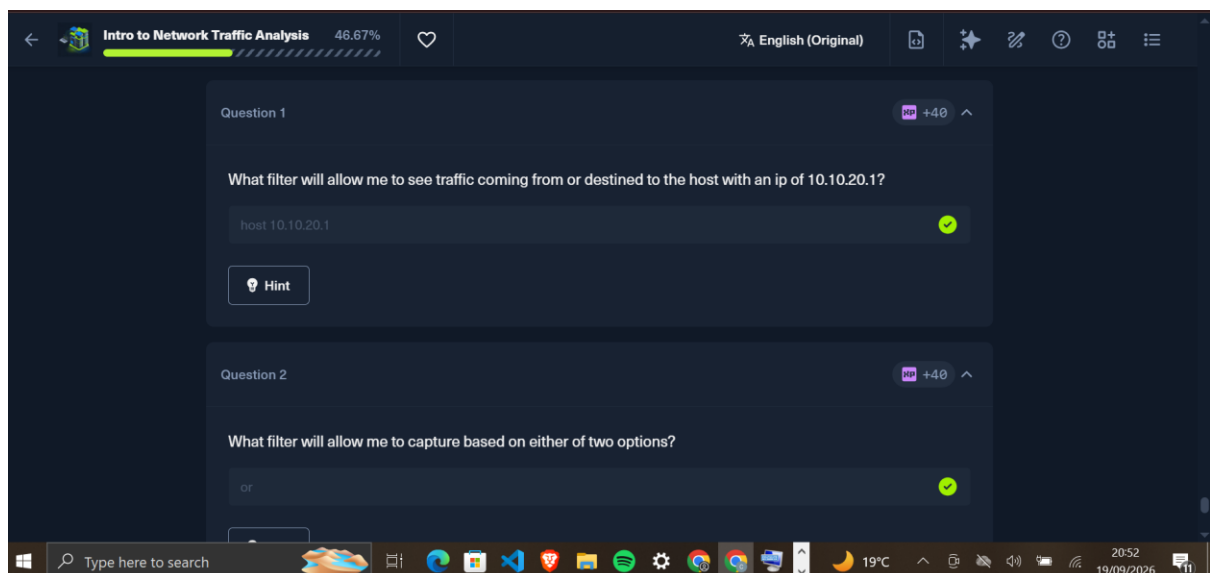
What filter will allow me to capture based on either of two options?

or

Question 3

True or False: TCPDump will resolve IPs to hostnames by default.

True



Week 2 Assignment 1 HTB Academy: Introduction to Network Traffic Analysis

Section: Interrogating Network Traffic With Capture and Display Filters

Question 1

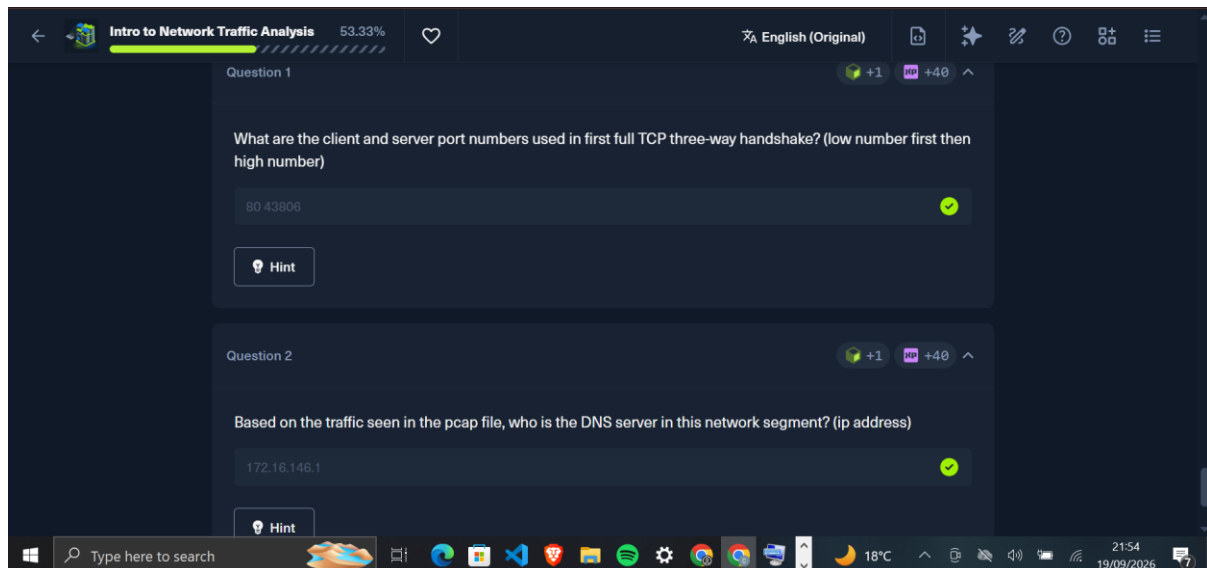
What are the client and server port numbers used in first full TCP three-way handshake? (low number first then high number)

80 43806

Question 2

Based on the traffic seen in the pcap file, who is the DNS server in this network segment? (ip address)

172.16.146.1



Module 4: Wireshark

Section: Analysis with Wireshark

Question 1

True or False: Wireshark can run on both Windows and Linux.

True

Question 2

Which Pane allows a user to see a summary of each packet grabbed during the capture?

packet list

Question 3

Which pane provides you insight into the traffic you captured and displays it in both ASCII and Hex?

packet bytes

Week 2 Assignment 1 HTB Academy: Introduction to Network Traffic Analysis

Question 4

What switch is used with TShark to list possible interfaces to capture on?

-D

Question 5

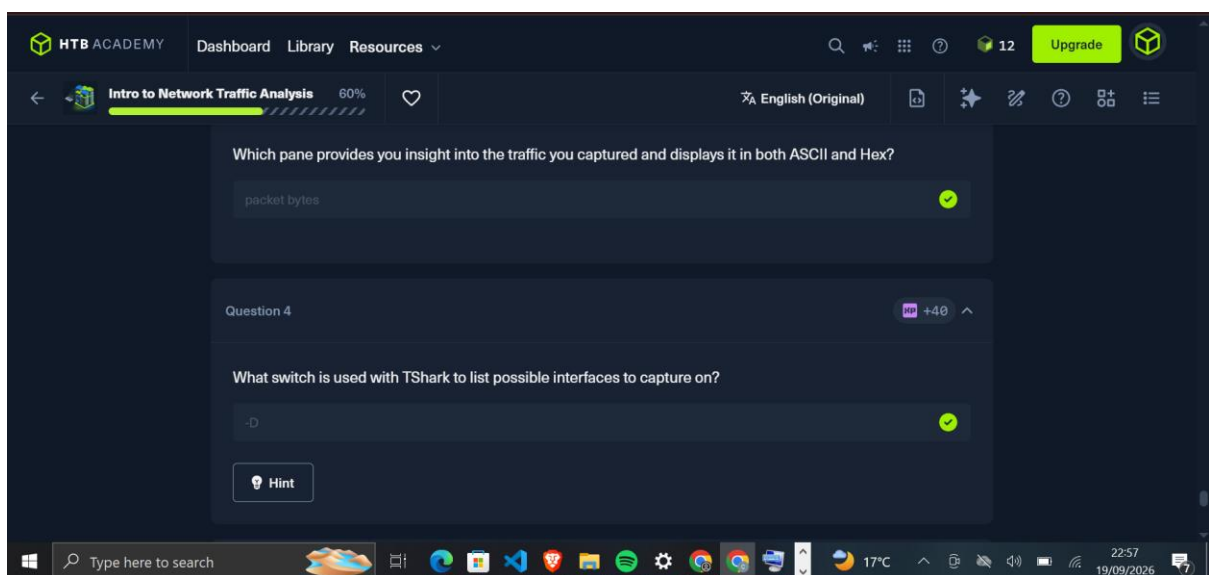
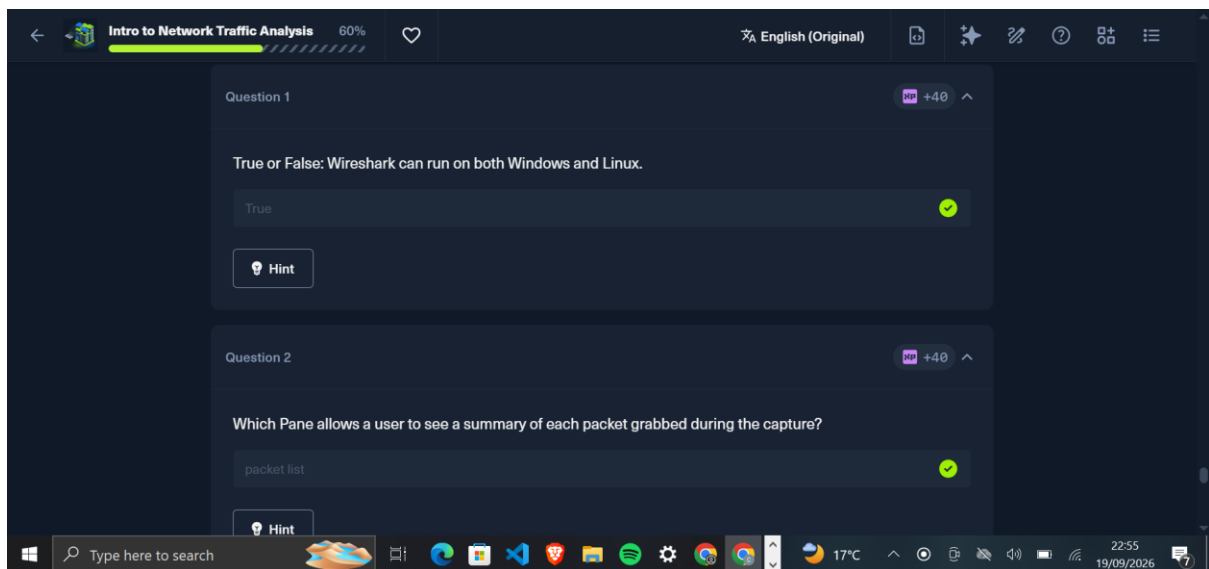
What switch allows us to apply filters in TShark?

-f

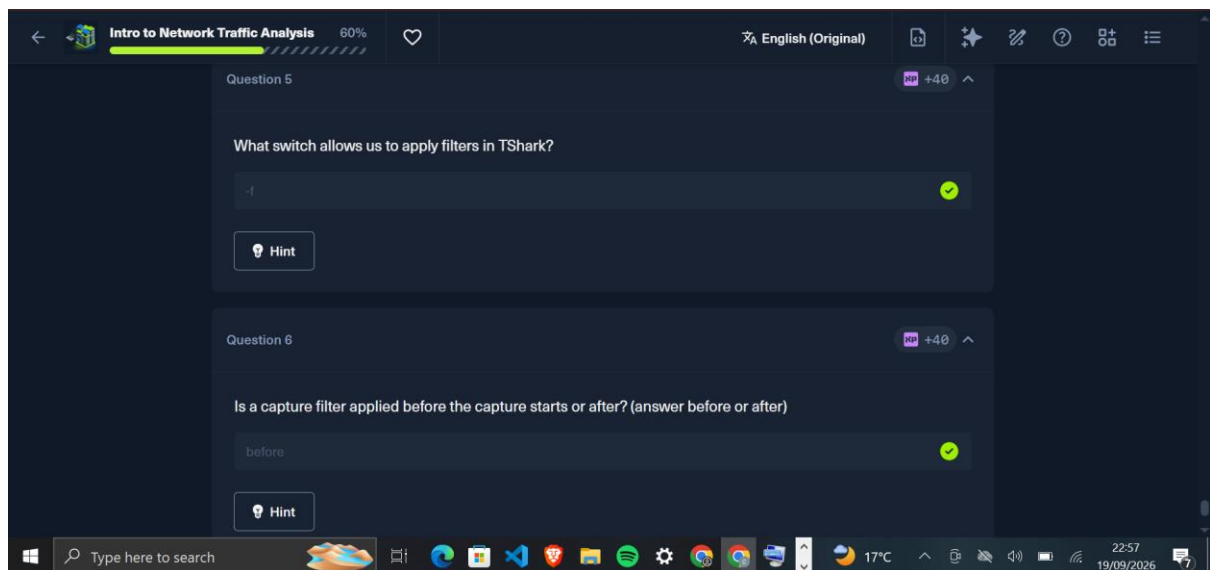
Question 6

Is a capture filter applied before the capture starts or after? (answer before or after)

before



Week 2 Assignment 1 HTB Academy: Introduction to Network Traffic Analysis



Section: Wireshark Advanced Usage

Question 1

Which plugin tab can provide us with a way to view conversation metadata and even protocol breakdowns for the entire PCAP file?

statistics

Question 2

What plugin tab will allow me to accomplish tasks such as applying filters, following streams, and viewing expert info?

analyze

Question 3

What stream oriented Transport protocol enables us to follow and rebuild conversations and the included data?

tcp

Question 4

True or False: Wireshark can extract files from HTTP traffic.

True

Question 5

True or False: The ftp-data filter will show us any data sent over TCP port 21.

False

Week 2 Assignment 1 HTB Academy: Introduction to Network Traffic Analysis

Intro to Network Traffic Analysis 73.33% English (Original)

Question 1

Which plugin tab can provide us with a way to view conversation metadata and even protocol breakdowns for the entire PCAP file?

statistics

Hint

Question 2

What plugin tab will allow me to accomplish tasks such as applying filters, following streams, and viewing expert info?

analyze

Type here to search

Live 00:12 20/09/2026

Intro to Network Traffic Analysis 73.33% English (Original)

What stream oriented Transport protocol enables us to follow and rebuild conversations and the included data?

tcp

Hint

Question 4

True or False: Wireshark can extract files from HTTP traffic.

True

Hint

Type here to search

Live 00:13 20/09/2026

Intro to Network Traffic Analysis 73.33% English (Original)

True

Hint

Question 5

True or False: The ftp-data filter will show us any data sent over TCP port 21.

False

Hint

Previous Section 12 / 15 Next Mark Complete & Next

Type here to search

Live 00:14 20/09/2026

Week 2 Assignment 1 HTB Academy: Introduction to Network Traffic Analysis

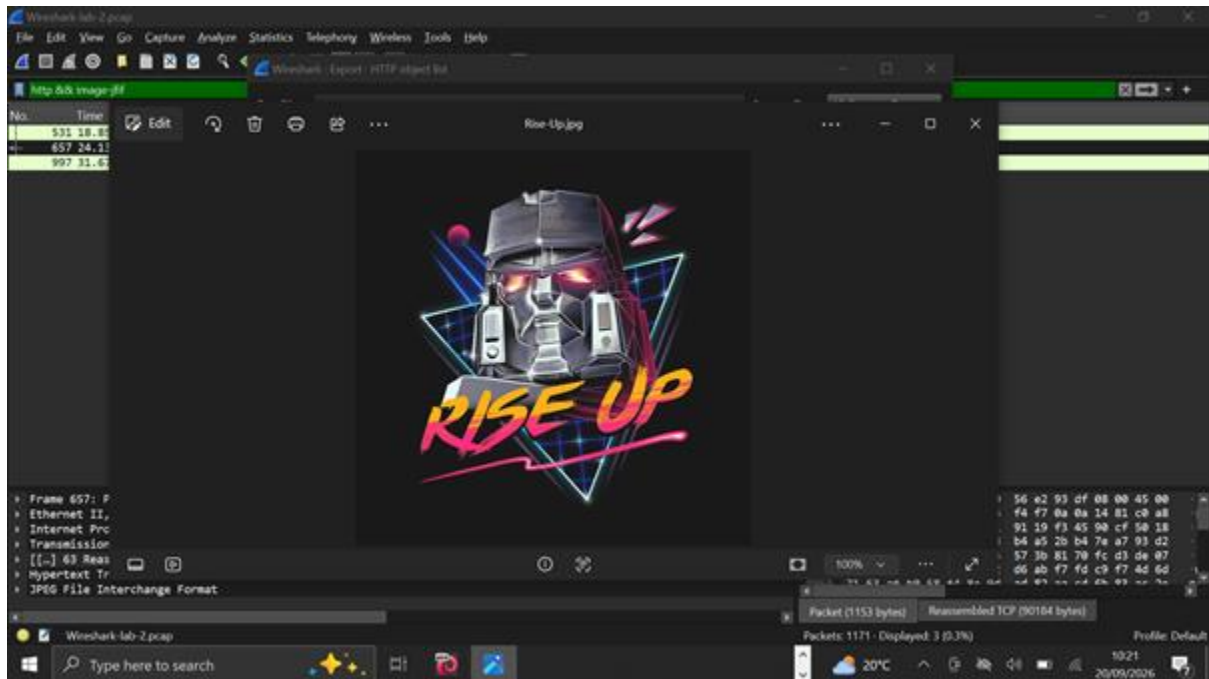
Section: Packet Inception, Dissecting Network Traffic With Wireshark

Question 1

What was the filename of the image that contained a certain Transformer Leader? (name.filetype)

RDP to 10.129.3.67 (ACADEMY-NTA-SNIFF01), with user "htb-student" and password "HTB_@cademy_stdnt!"

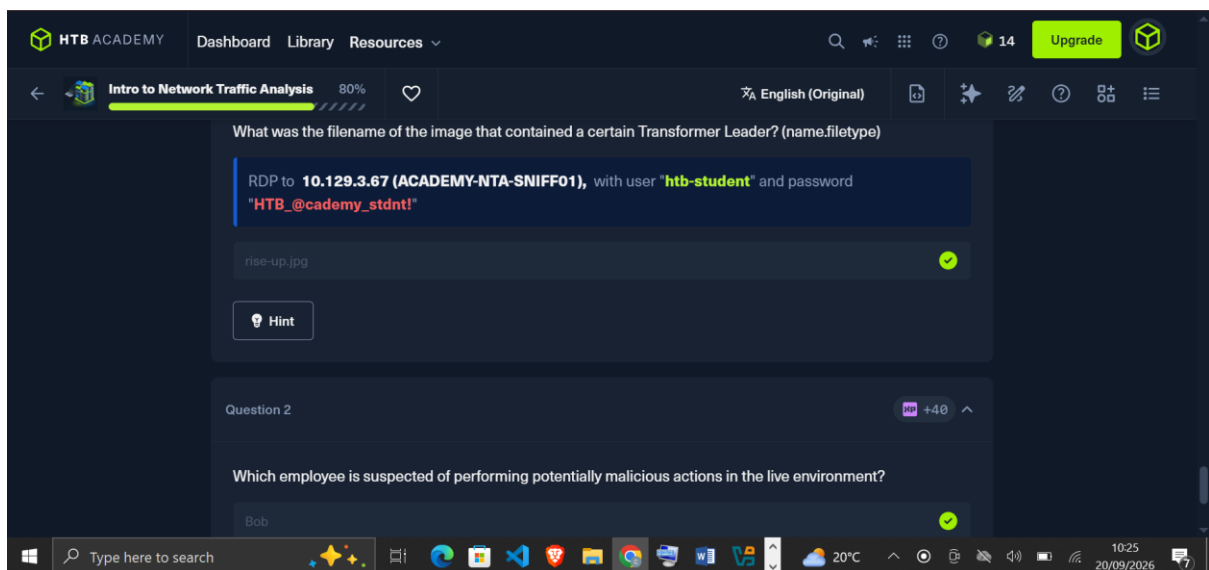
rise-up.jpg



Question 2

Which employee is suspected of performing potentially malicious actions in the live environment?

Bob



Week 2 Assignment 1 HTB Academy: Introduction to Network Traffic Analysis

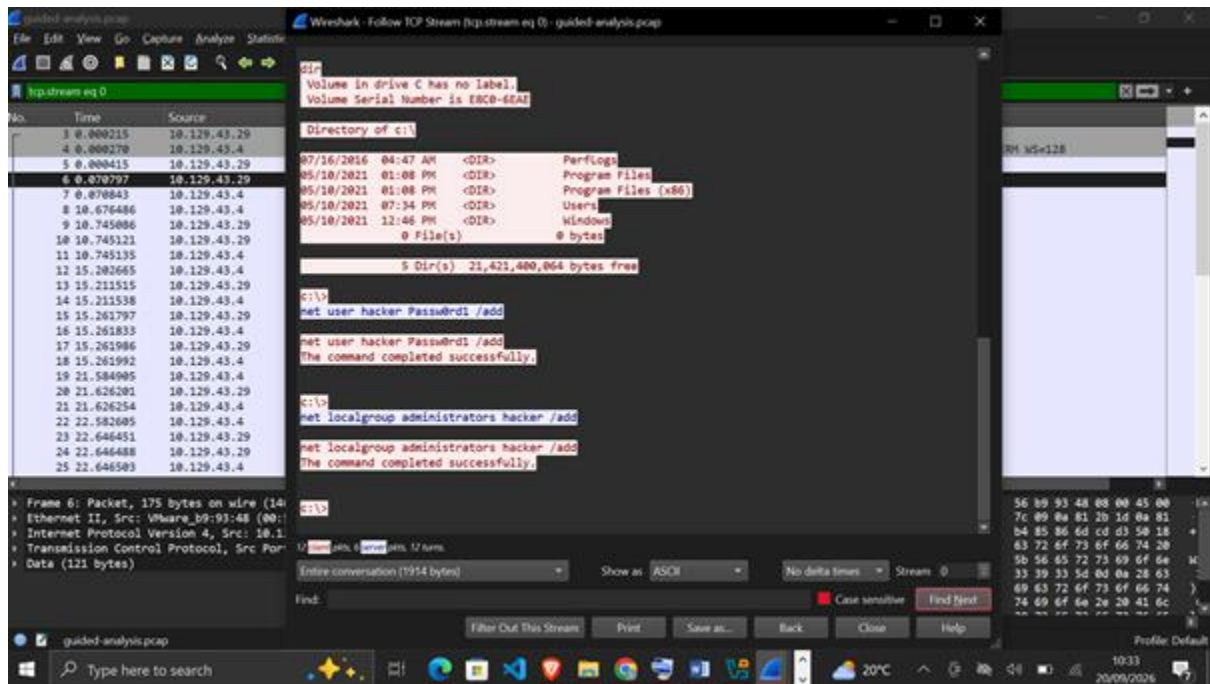
Section: Guided Lab- Traffic Analysis Workflow

Question 1

What was the name of the new user created on mrb3n's host?

RDP to 10.129.3.67 (ACADEMY-NTA-SNIFF01), with user "htb-student" and password "HTB_@cademy_stdnt!"

Hacker



Question 2

How many total packets were there in the Guided-analysis PCAP?

44

Question 3

What was the suspicious port that was being used?

4444

Week 2 Assignment 1 HTB Academy: Introduction to Network Traffic Analysis

The image shows a Wireshark packet capture analysis of a TCP stream. The top pane displays a list of packets, with packet 6 selected. The middle pane shows the details of packet 6, highlighting the TCP segment. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
3	0.000215	10.129.43.29	10.129.43.4	TCP	66	50612 → 4444 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM
4	0.000270	10.129.43.4	10.129.43.29	TCP	66	4444 → 50612 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
5	0.000415	10.129.43.29	10.129.43.4	TCP	60	50612 → 4444 [ACK] Seq=1 Ack=1 Win=2102272 Len=0
6	0.070797	10.129.43.29	10.129.43.4	TCP	175	50612 → 4444 [PSH, ACK] Seq=1 Ack=1 Win=2102272 Len=121

Frame 6: Packet, 175 bytes on wire (1400 bits), 175 bytes captured (1400 bits) on interface 0
Ethernet II, Src: VMware_b9:93:48 (00:50:56:b9:93:48), Dst: VMware_b9:6c:2c (00:50:56:b9:6c:2c)
Internet Protocol Version 4, Src: 10.129.43.29, Dst: 10.129.43.4
Transmission Control Protocol, Src Port: 50612, Dst Port: 4444, Seq: 1, Ack: 1, Len: 121
Data (121 bytes)

The image shows a quiz interface for 'Intro to Network Traffic Analysis'. The quiz consists of two questions. The first question asks for the name of the new user created on mrb3n's host. The second question asks for the total number of packets in the Guided-analysis PCAP.

Question 1: What was the name of the new user created on mrb3n's host?

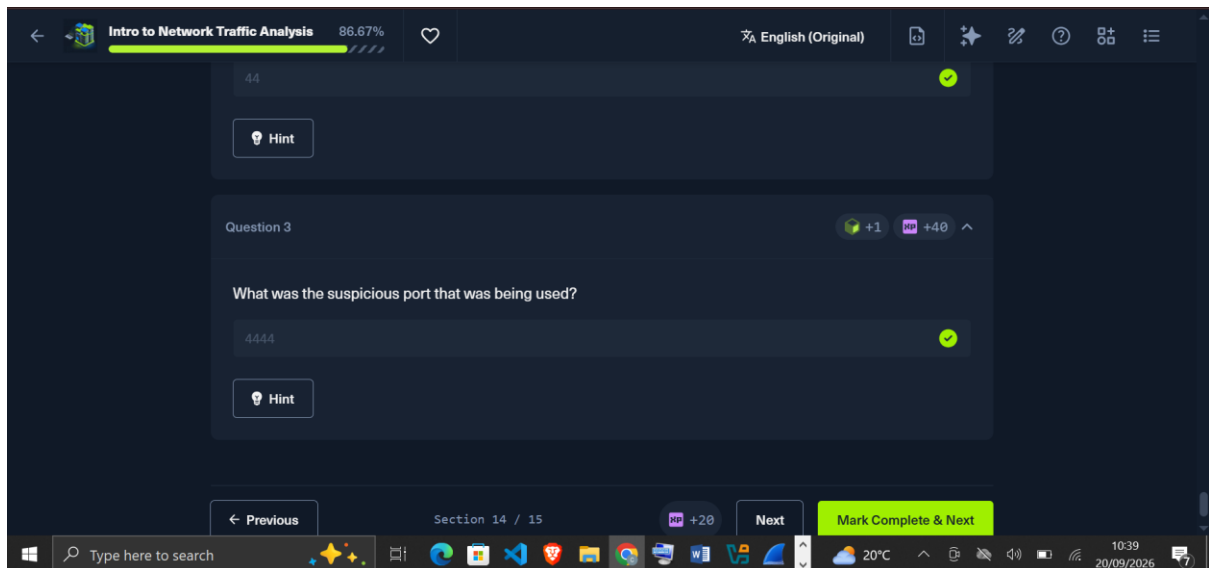
RDP to 10.129.3.67 (ACADEMY-NTA-SNIF01), with user "htb-student" and password "HTB@cademy_stdnt!"

hacker

Question 2: How many total packets were there in the Guided-analysis PCAP?

44

Week 2 Assignment 1 HTB Academy: Introduction to Network Traffic Analysis

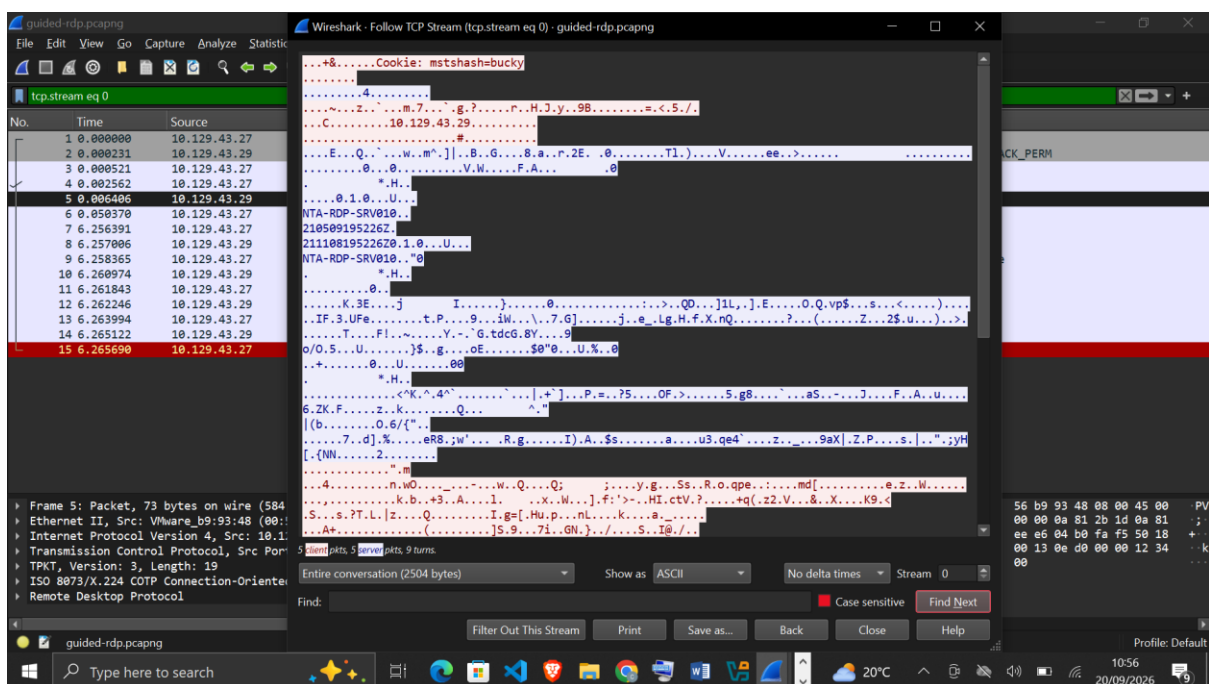


Section: Decrypting RDP connections

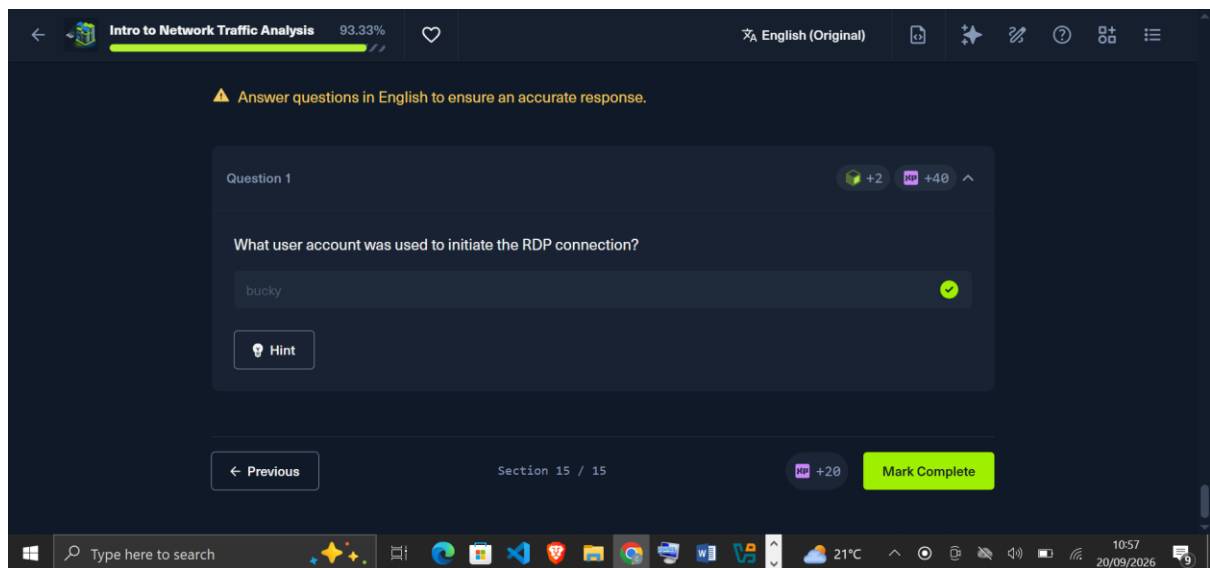
Question 1

What user account was used to initiate the RDP connection?

ucky



Week 2 Assignment 1 HTB Academy: Introduction to Network Traffic Analysis

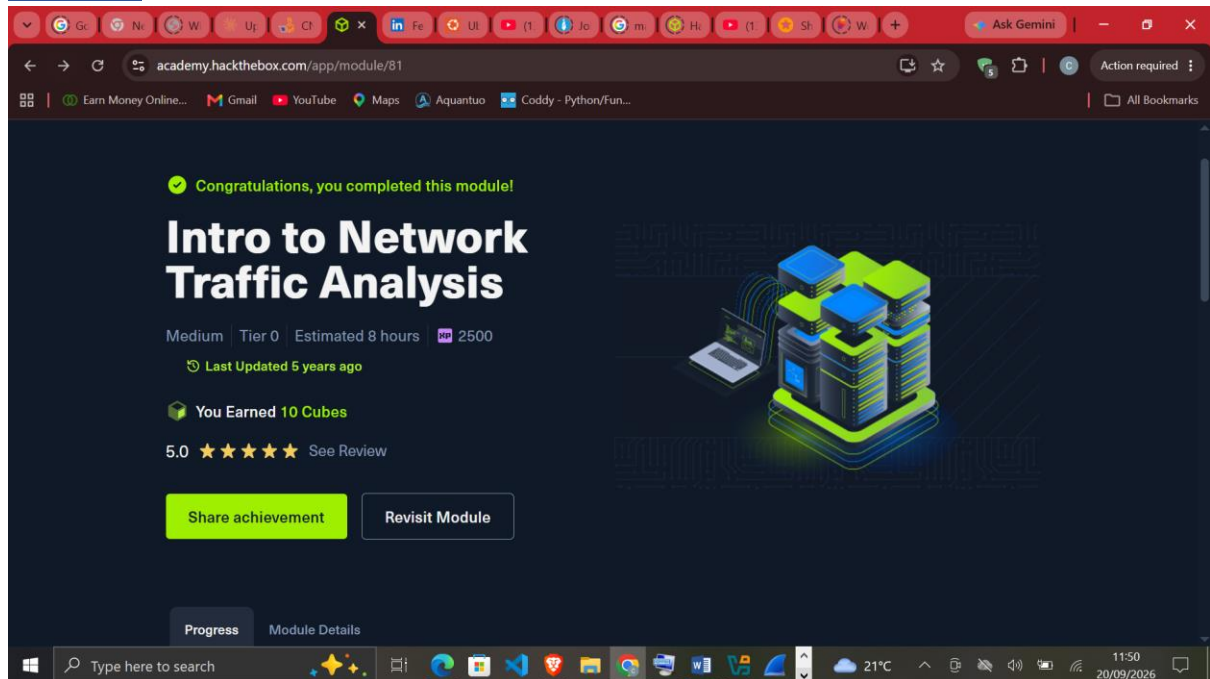


Conclusion

This activity really outlined the fundamentals of Network Traffic Analysis as a foundational skill for security professionals. It bridged the gap between theoretical models and real-world system behaviour. By mastering packet inspection tools like **tcpdump**, **TShark**, and **Wireshark**, you can peel back network encapsulation, reconstruct protocol flows, trace active connections, and spot anomalies across all layers of communication. Furthermore, hands-on techniques such as accessing remote hosts via **xfreerdp** enable real-time packet capturing and forensic investigation directly inside target environments. Applying these access methods alongside packet inspection and filtering workflows will allow you to monitor live traffic, investigate security incidents, and systematically defend your network infrastructure.

Week 2 Assignment 1 HTB Academy: Introduction to Network Traffic Analysis

Appendix



1 Completion Screenshot

Completion link:

<https://academy.hackthebox.com/achievement/2433236/81>